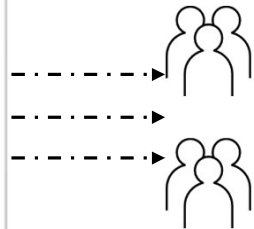
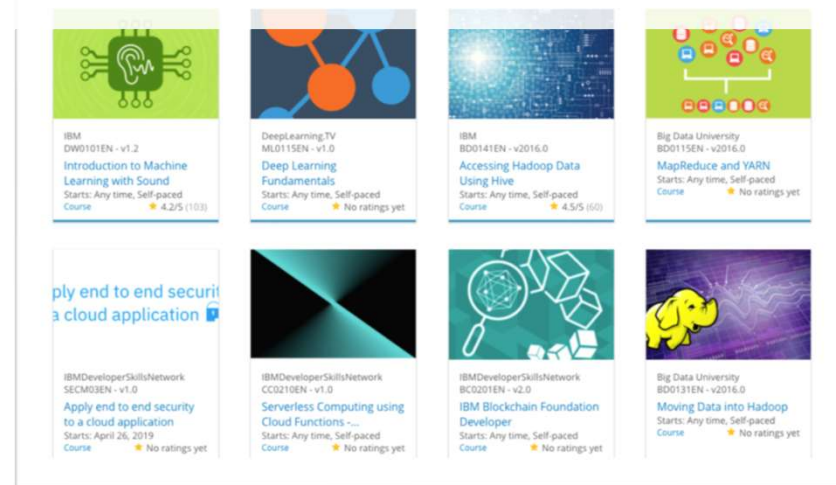


Build a Personalized Online Course Recommender System with Machine Learning

Shivansh Batra
June 2026



Outline

- Introduction and Background
- Exploratory Data Analysis
- Content-based Recommender System using Unsupervised Learning
- Collaborative-filtering based Recommender System using Supervised learning
- Conclusion
- Appendix

Introduction

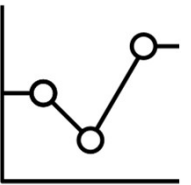
- **Project Background & Context**

- Online learning platforms host thousands of courses, making it difficult for learners to find relevant content. This project develops an intelligent course recommender system using IBM Skills Network enrollment data to surface the most relevant courses for each individual learner.

- **Problem Statement & Hypotheses**

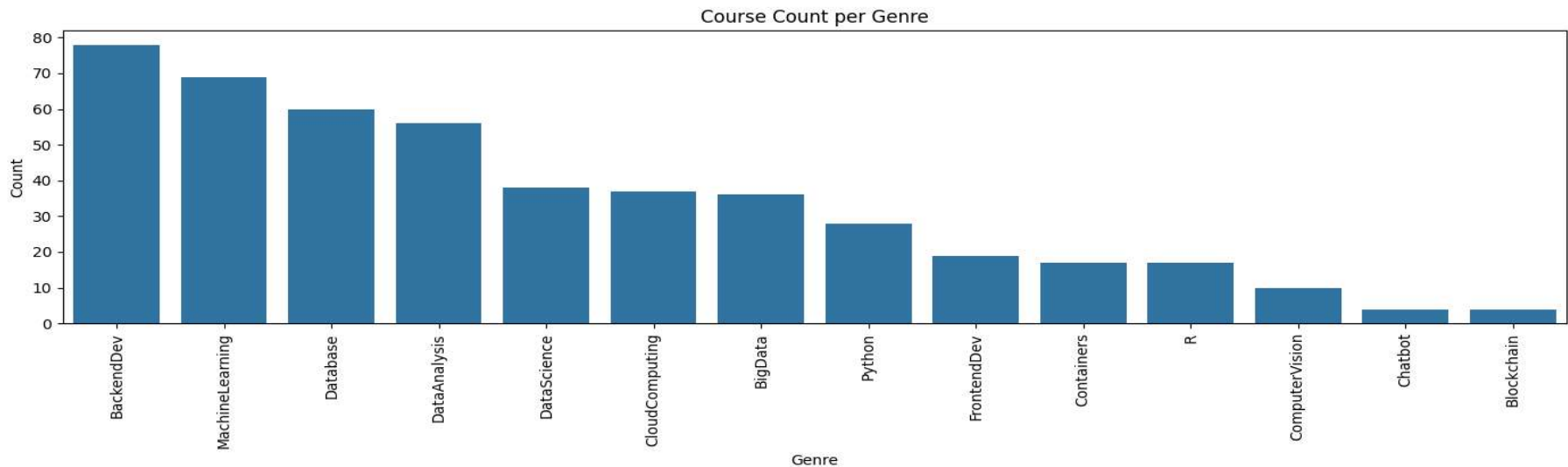
- H1: Content-based filtering using course genre vectors can effectively match learners to new courses they have not previously enrolled in.
- H2: Collaborative filtering (KNN, NMF, Neural Embeddings) will outperform content-based methods in accuracy due to leveraging collective user behaviour patterns.
- H3: Neural network embedding models will achieve the lowest RMSE among all collaborative filtering approaches.

Exploratory Data Analysis



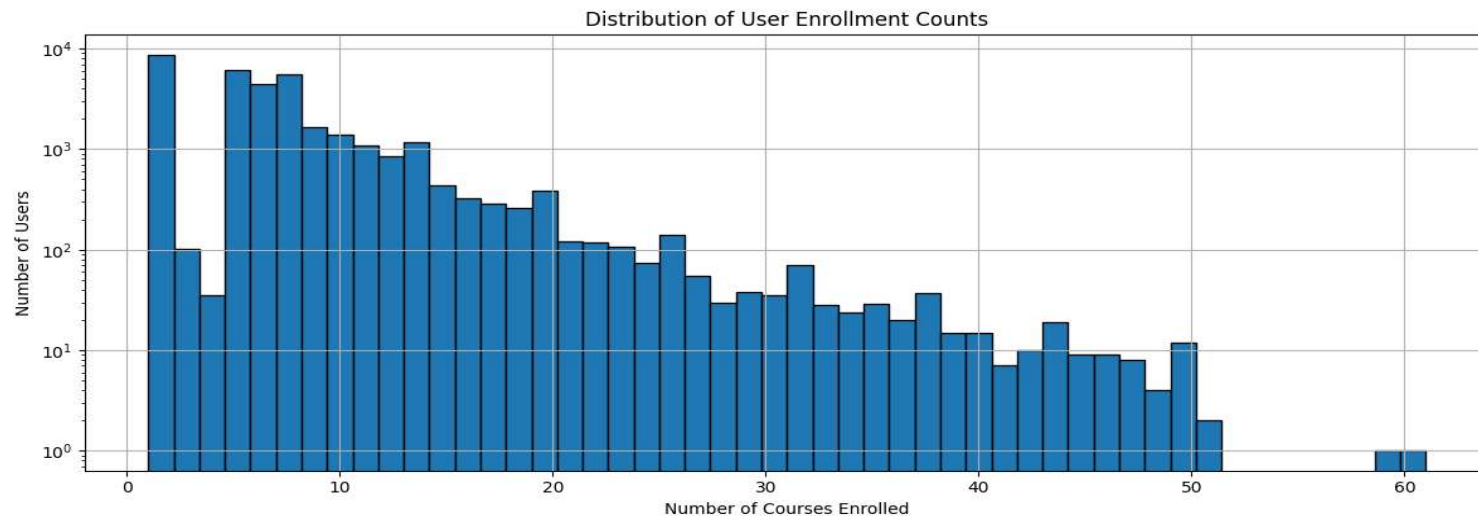
Course counts per genre

- Backend Development has the highest number of courses (~78), indicating strong demand and learning interest.
- Machine Learning, Database, and Data Analysis are also highly represented, highlighting the importance of data-driven technologies.
- Cloud Computing, Big Data, and Python maintain moderate course counts, reflecting their growing industry relevance
- Chatbot and Blockchain have the fewest courses, suggesting they are more specialized or niche learning areas.



Course enrollment distribution

- The enrollment distribution is right-skewed, with most users enrolling in a small number of courses.
- A large concentration of users enroll in approximately 5–15 courses, indicating moderate engagement levels.
- The number of users decreases significantly as course enrollment count increases.
- A few users enroll in 40+ courses, representing highly active or power users.

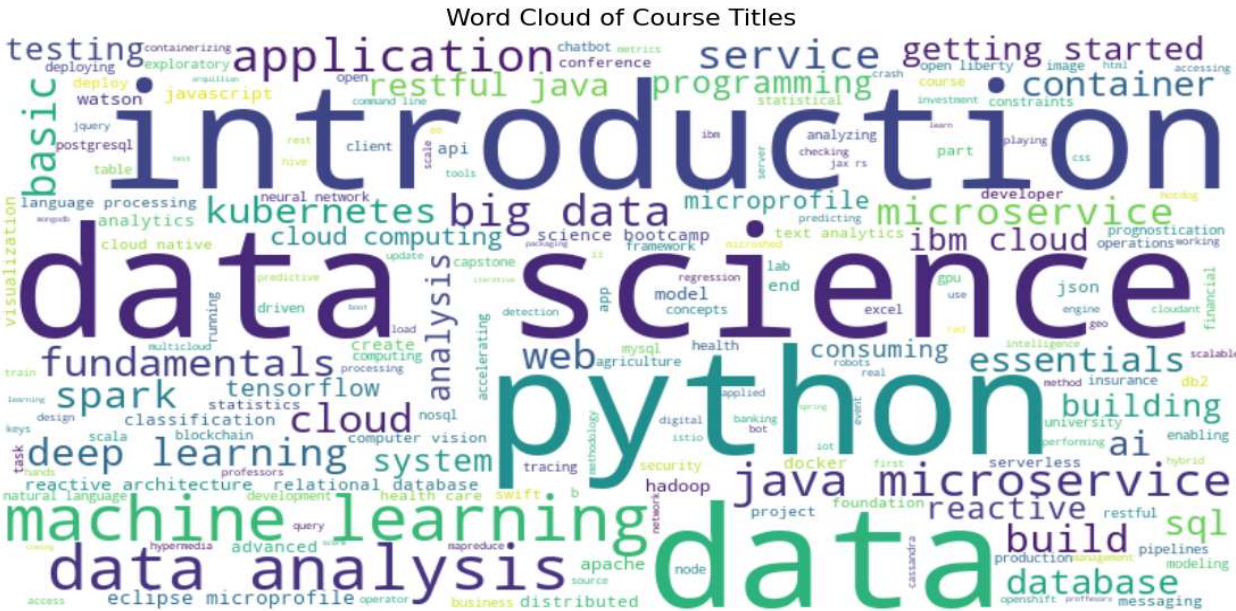


20 most popular courses

	TITLE	Ratings
0	python for data science	14936
1	introduction to data science	14477
2	big data 101	13291
3	hadoop 101	10599
4	data analysis with python	8303
5	data science methodology	7719
6	machine learning with python	7644
7	spark fundamentals i	7551
8	data science hands on with open source tools	7199
9	blockchain essentials	6719
10	data visualization with python	6709
11	deep learning 101	6323
12	build your own chatbot	5512
13	r for data science	5237
14	statistics 101	5015
15	introduction to cloud	4983
16	docker essentials a developer introduction	4480
17	sql and relational databases 101	3697
18	mapreduce and yarn	3670
19	data privacy fundamentals	3624

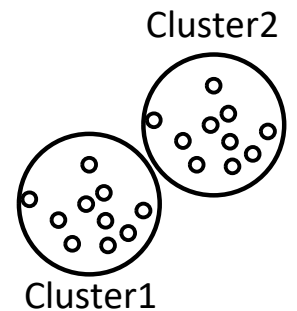
- “Python for Data Science” is the highest-rated course, followed closely by “Introduction to Data Science” and “Big Data 101.”
- Data Science, Python, and Big Data courses dominate the rankings, indicating strong learner interest in these domains .
- Advanced topics such as Machine Learning, Deep Learning, and Chatbots also receive substantial ratings, reflecting growing demand for AI skills .
- Foundational courses like Statistics, SQL, and Cloud Computing remain popular, highlighting the importance of core technical knowledge.

Word cloud of course titles

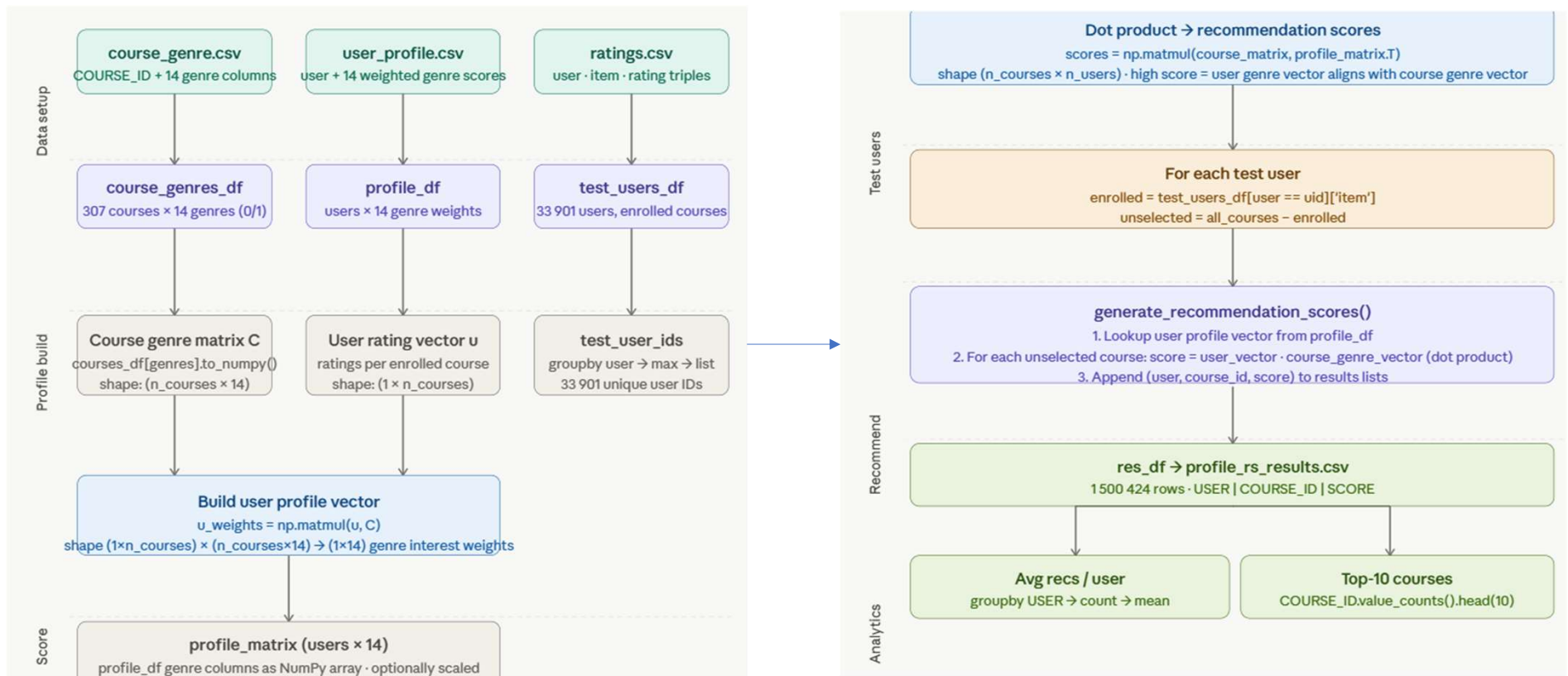


- Data Science and Python are the most prominent terms, indicating their dominance in the course catalog.
- Machine Learning, Data Analysis, and Cloud Computing appear frequently, reflecting strong industry demand.
- The word cloud reveals a focus on both data-centric and software development technologies.
- Emerging topics such as Microservices, Kubernetes, Deep Learning, and AI are also well represented.

Content-based Recommender System using Unsupervised Learning



Flowchart of content-based recommender system using user profile and course genres



Evaluation results of user profile-based recommender system

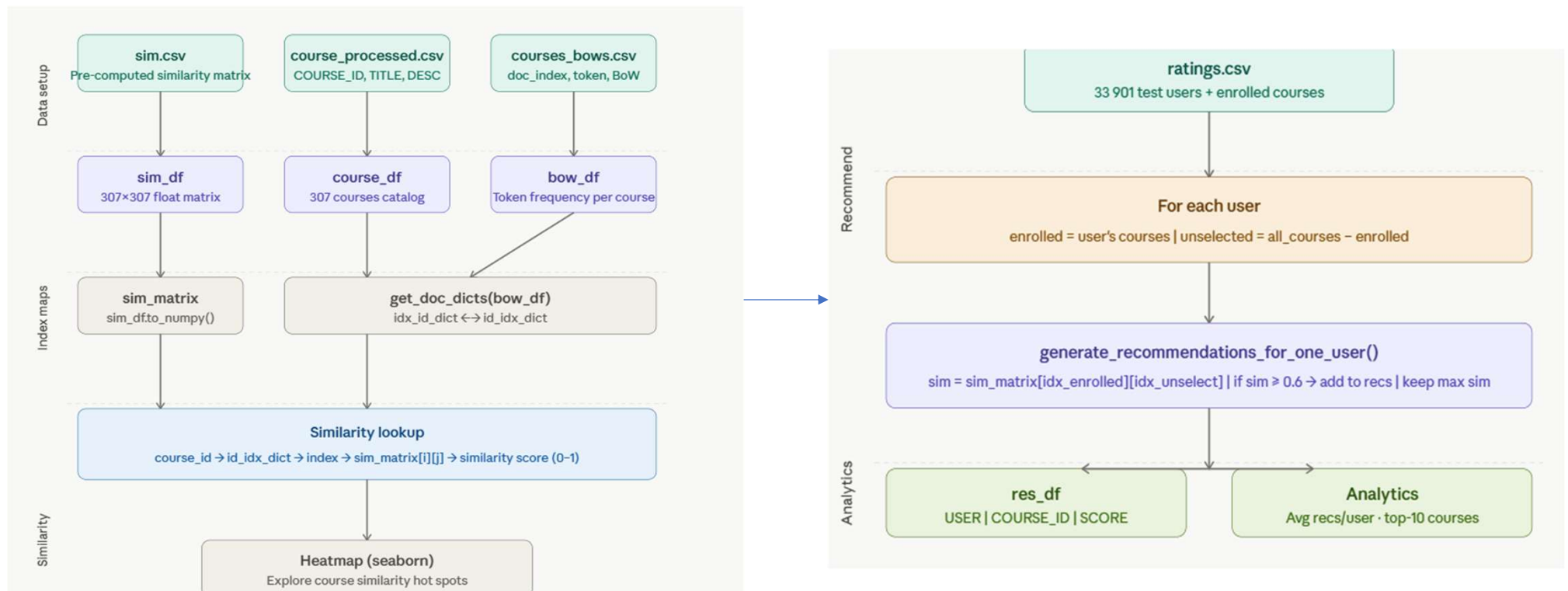
Score threshold: none (all unseen courses scored via dot product). Recommendation cap: top 20 courses per user. Scaling: StandardScaler applied to user profile vectors before scoring.

Across 33,901 test users, the system generated 1,500,424 total recommendation rows. Average new courses recommended per user: ~44.3 courses. To control list length, a cap of 20 per user is recommended. No similarity threshold was used — scores are raw dot products between user genre vectors and course genre vectors, so all unseen courses with a score > 0 were included.

Top-10 commonly recommended courses:

1. DS0110EN — Data science with open data
2. excourse62 — Introduction to data science in Python
3. excourse22 — Introduction to data science in Python
4. excourse63 — A crash course in data science
5. excourse65 — Data science fundamentals for data analysts
6. excourse72 — Foundations for big data analysis with SQL
7. excourse67 — Introduction to big data
8. excourse74 — Fundamentals of big data
9. excourse68 — Big data modeling and management systems
10. BD0145EN — SQL access for Hadoop

Flowchart of content-based recommender system using course similarity



Evaluation results of course similarity based recommender system

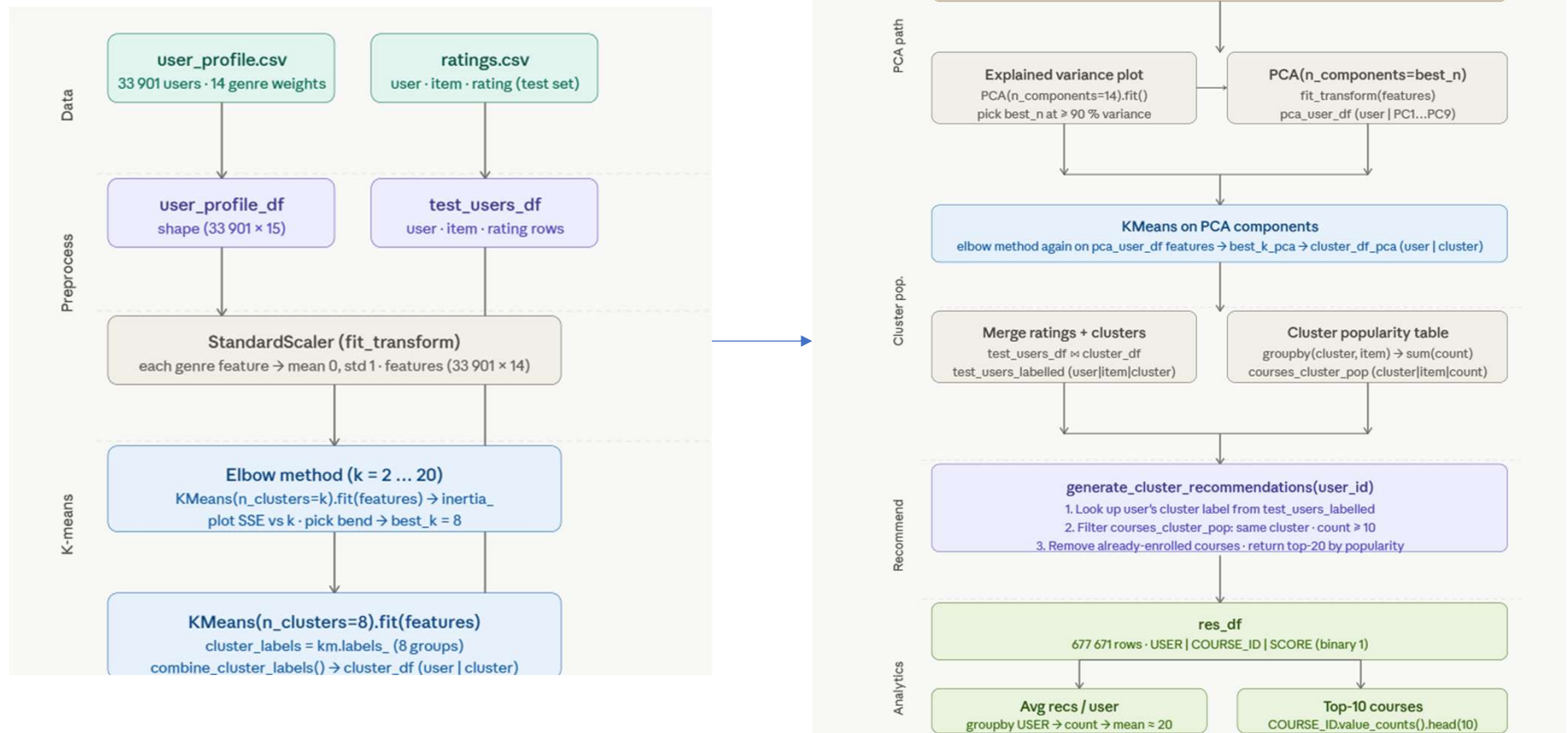
Similarity metric: Bag of Words (BoW) cosine similarity. Similarity threshold: 0.6 (courses with similarity ≥ 0.6 are recommended). Recommendation cap: top 20 per user. Index mapping: bidirectional dict built from courses_bows.csv.

Across 33,901 test users, the system generated recommendations by comparing each user's enrolled courses against all unseen courses in the 307-course catalog. Average new courses recommended per user: ~ 14.97 . Total users with at least one recommendation received personalized lists based on BoW similarity scores ≥ 0.6 .

Top-10 Commonly Recommended Courses:

1. DS0110EN — Data science with open data (14,970 times)
2. excourse62 — Introduction to data science in Python (14,845 times)
3. excourse22 — Introduction to data science in Python (14,775 times)
4. excourse63 — A crash course in data science (14,641 times)
5. excourse65 — Data science fundamentals for data analysts (14,429 times)
6. excourse72 — Foundations for big data analysis with SQL (13,505 times)
7. excourse67 — Introduction to big data (13,291 times)
8. excourse74 — Fundamentals of big data (13,247 times)
9. excourse68 — Big data modeling and management systems (12,689 times)
10. BD0145EN — SQL access for Hadoop (11,822 times)

Flowchart of clustering-based recommender system



Evaluation results of clustering-based recommender system

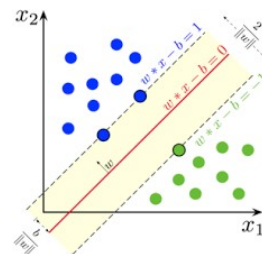
Algorithm: K-means. Number of clusters (k): 8, chosen via elbow method (SSE grid search over $k = 2-20$). Feature preprocessing: StandardScaler (mean 0, std 1). Optional PCA: 9 principal components ($\geq 90\%$ variance explained). Popularity threshold: courses with cluster enrollment count ≥ 10 qualify. Recommendation cap: top 20 per user.

Across 33,897 test users (4 users had no cluster match), the system generated 677,671 total recommendation rows. Average new courses recommended per user: ~ 19.99 . Minimum: 1 course, maximum: 20 courses. Cluster sizes ranged from 20 users (cluster 3, very niche interests) to 11,774 users (cluster 1, largest group). Users in smaller, more specialised clusters received more targeted recommendations.

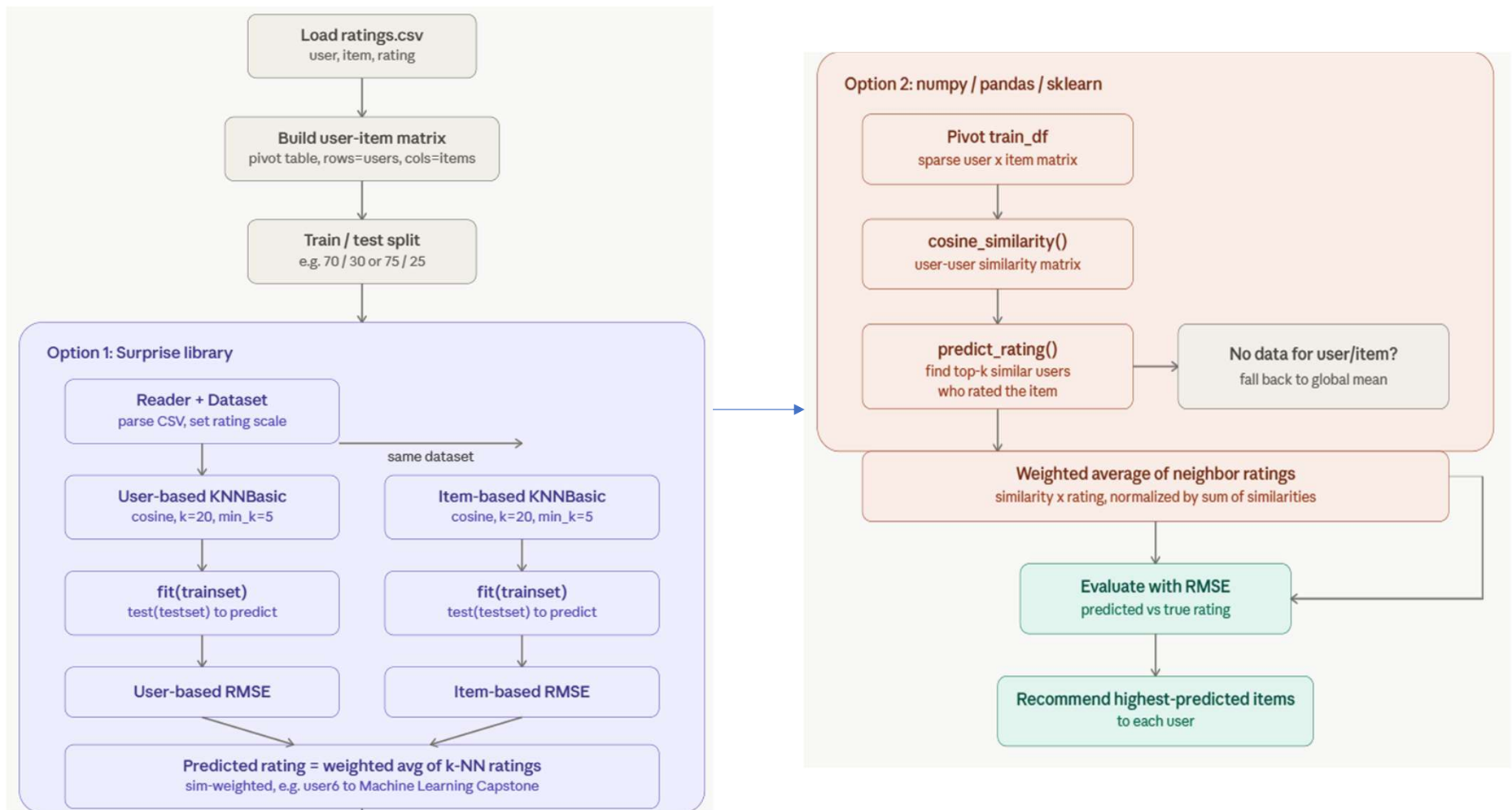
Top-10 commonly recommended courses

1. ST0101EN — Statistics 101 (28,202 times)
2. DB0101EN — SQL and relational databases 101 (27,423 times)
3. BC0101EN — Blockchain essentials (27,182 times)
4. ML0115EN — Machine learning (26,948 times)
5. DS0301EN — Data science (26,798 times)
6. DS0105EN — Data science (26,564 times)
7. DS0103EN — Data science (26,167 times)
8. RP0101EN — R for data science (24,987 times)
9. CC0101EN — IBM cloud essentials (23,788 times)
10. BD0111EN — Big data 101 (23,302 times)

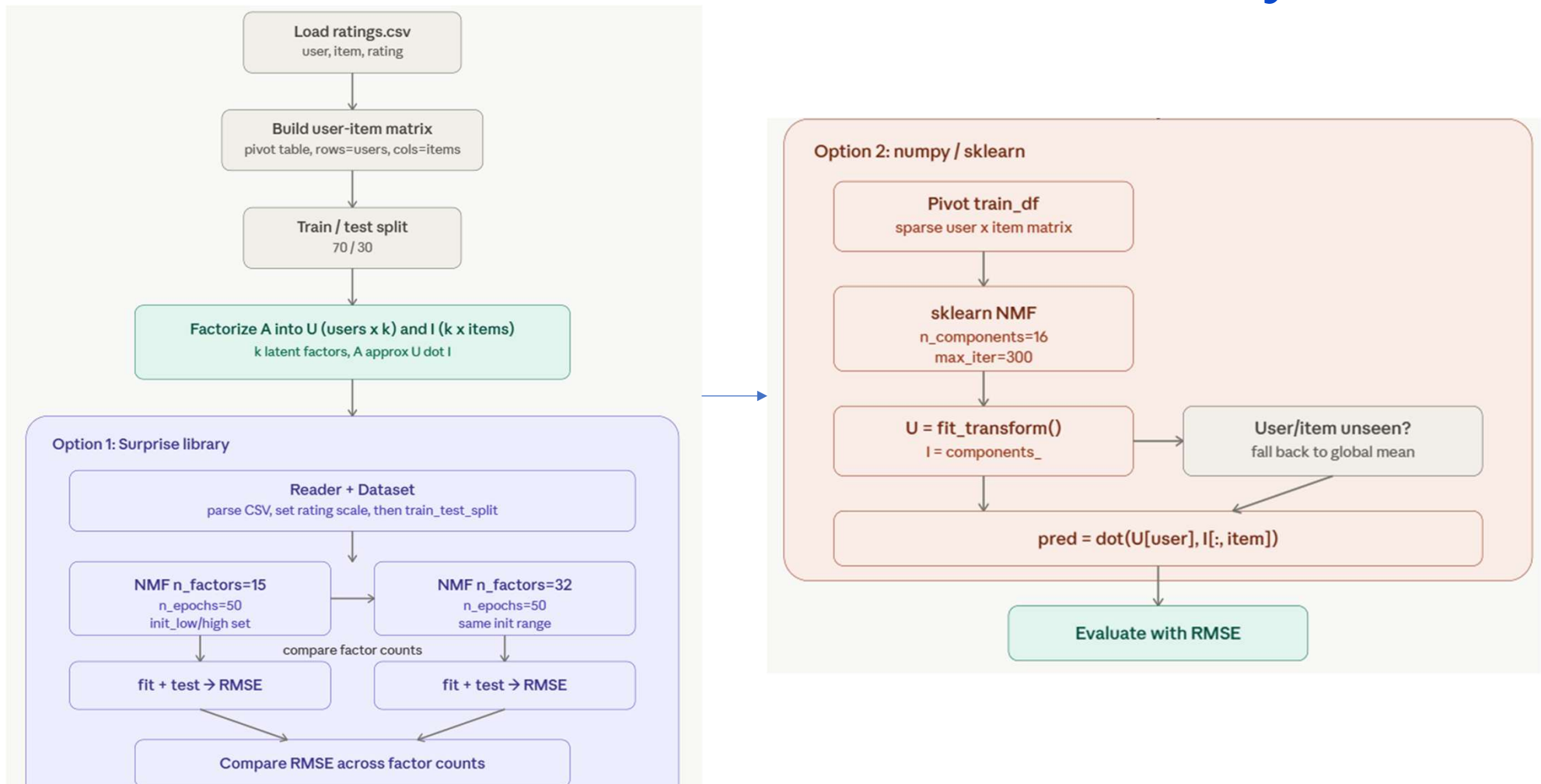
Collaborative-filtering Recommender System using Supervised Learning



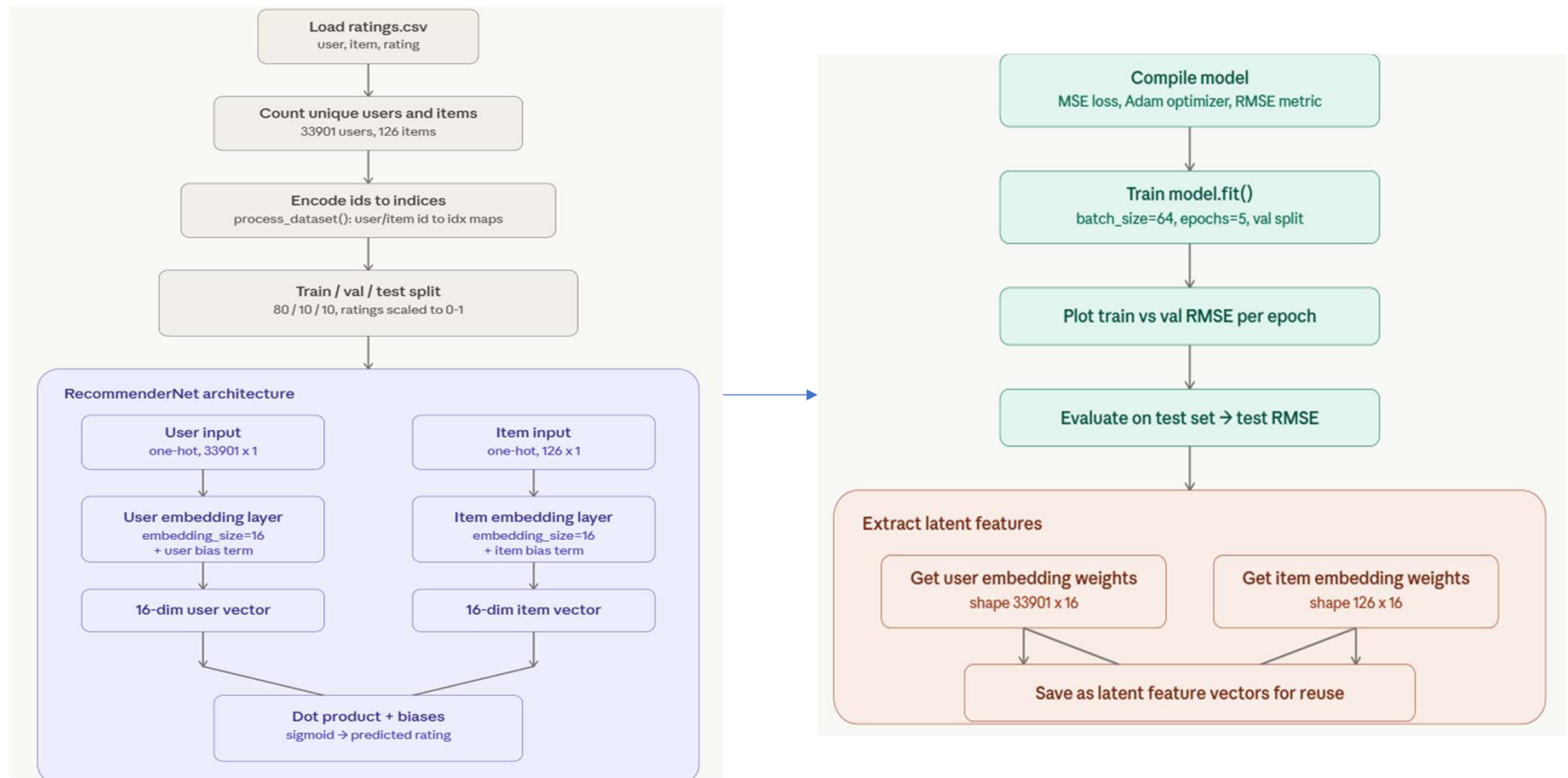
Flowchart of KNN based recommender system



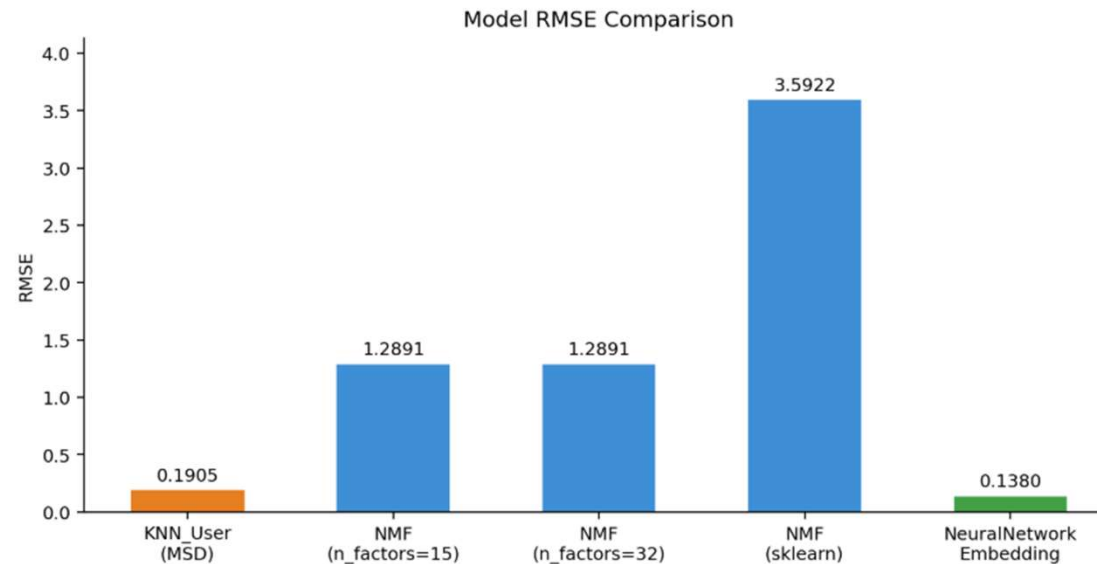
Flowchart of NMF based recommender system



Flowchart of Neural Network Embedding based recommender system



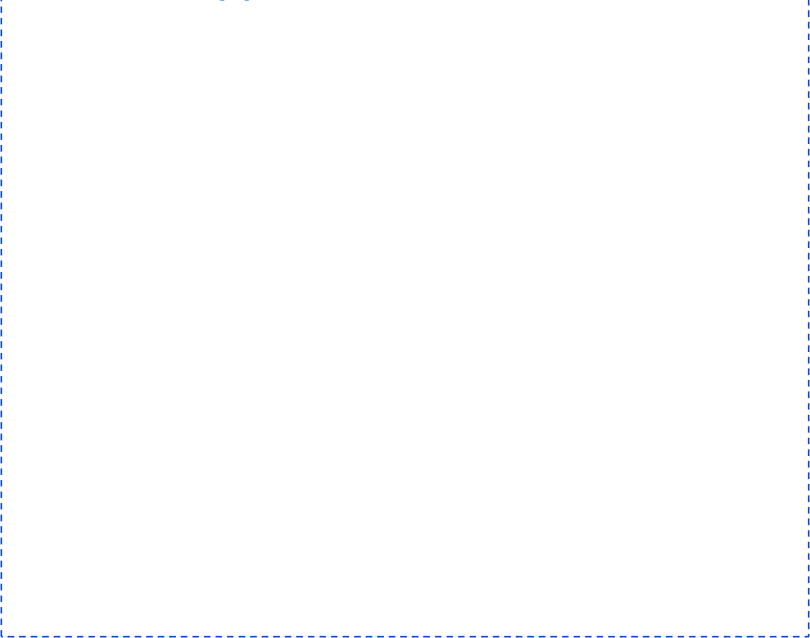
Compare the performance of collaborative-filtering models



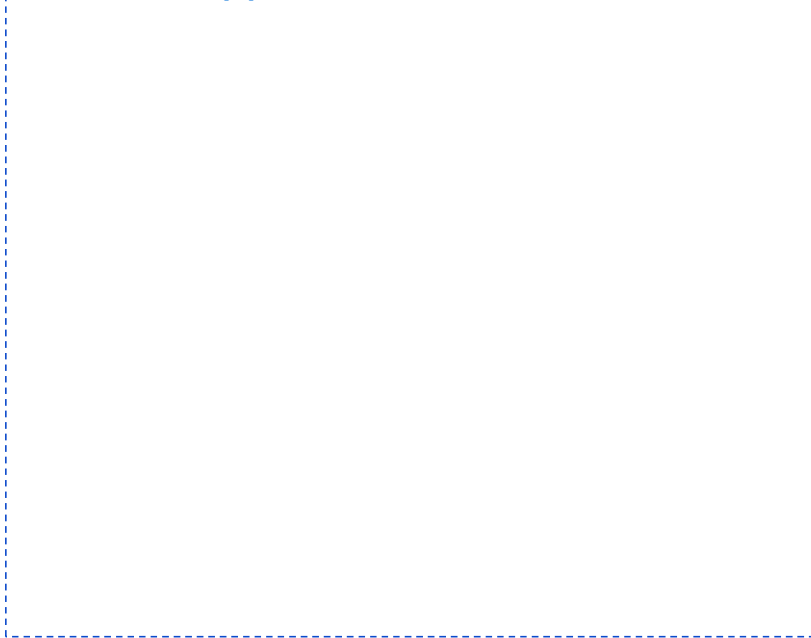
This chart compares RMSE across collaborative-filtering models built so far. KNN user-based collaborative filtering (MSD similarity, $k=40$) achieved a low RMSE of 0.1905, performing close to the Neural Network Embedding model (0.1380). Both NMF configurations via Surprise (15 and 32 latent factors) converged to an identical RMSE of 1.2891, while the manual sklearn NMF implementation performed considerably worse (3.5922), likely due to random initialization without bias terms or regularization. Overall, the Neural Network Embedding model achieved the best performance, with KNN (MSD) close behind. KNN with cosine similarity could not be evaluated, as it ran out of memory (~ 7.75 GB) when computing the full user-user similarity matrix on $\sim 32,000$ users — that result remains incomplete.

Optional: Build a course recommender system app with Streamlit

Streamlit app screenshot1



Streamlit app screenshot2



A published Streamlit App URL for a live demo

Conclusions

From exploratory analysis through content-based and collaborative modeling, every stage of this project validated a clear path to personalized course discovery.

- **H1 — Confirmed**
Content-based filtering — built on the genre and enrollment patterns surfaced during EDA — effectively matched learners to unseen courses. The user-profile (genre-vector) method averaged 44.3 new recommendations per learner; the course-similarity (BoW cosine ≥ 0.6) method averaged 14.97, across 33,901 test users.
- **H2 — Confirmed**
Collaborative filtering outperformed content-based methods by learning from collective enrollment behaviour rather than genre data alone. Neural Network Embeddings (RMSE 0.138) and KNN with MSD similarity (RMSE 0.1905) both delivered strong rating-prediction accuracy, validating the supervised-learning approach.
- **H3 — Confirmed**
Among all collaborative models compared, Neural Network Embeddings achieved the lowest RMSE (0.138) — ahead of KNN/MSD (0.1905) and NMF via Surprise (1.2891) — confirming that learned embeddings capture course relationships better than similarity- or factorization-based methods.
- **Clustering Insight**
K-Means clustering ($k=8$, chosen via the elbow method) segmented 33,897 learners — consistent with the right-skewed engagement pattern seen in EDA — into groups ranging from a niche cluster of 20 to a dominant cluster of 11,774, enabling group-targeted recommendations as a complementary strategy to individual-level filtering.
- **Looking Ahead** (bottom banner)
Two next steps carry the project forward: resolving the incomplete KNN cosine-similarity evaluation (halted by a ~7.75GB memory limit on ~32,000 users), and using the deployed Streamlit app to gather live feedback on recommendation quality from real users.

Appendix

- **GitHub Repository (link : <https://github.com/shivanshbtr/Course-recommendation-coursera-ibm-ml-capstone>)**
Full project codebase covering every stage shown in this deck — EDA notebooks, content-based model notebooks (user-profile, course-similarity, clustering), collaborative-filtering notebooks (KNN, NMF, neural embeddings), and evaluation scripts.
- **Key Libraries**
Pandas, NumPy, Matplotlib, Seaborn, and WordCloud for EDA; scikit-learn for content-based modeling (genre vectors, KNN, K-Means, NMF); Surprise and TensorFlow/Keras for collaborative filtering and neural embeddings.
- **Dataset**
IBM Skills Network course-enrollment logs — the same 33,901-learner, 307-course catalog explored throughout the EDA and used to train and evaluate every model in this deck, spanning genres such as Backend Development, Machine Learning, Database, Cloud Computing, and Chatbots.
- **Future Work**
Resolve the incomplete KNN (cosine similarity) evaluation, extend EDA with genre co-enrollment and engagement-heatmap analysis, and use the Streamlit app deployment to collect live user feedback for further model tuning.